

Time-independent Schrodinger equation

Let $\Psi(x, t)$ be the wave function

$$\Psi(x, t) = \psi(x) \exp\left(-\frac{i}{\hbar}Et\right)$$

For the time derivative we have

$$i\hbar \frac{d}{dt} \Psi(x, t) = E\Psi(x, t) \quad (1)$$

Recall the Schrodinger equation

$$i\hbar \frac{d}{dt} \Psi(x, t) = H\Psi(x, t) \quad (2)$$

By the equivalence of (1) and (2)

$$E\Psi(x, t) = H\Psi(x, t)$$

Let H be independent of t . Then by linearity of H

$$H\Psi(x, t) = H\psi(x) \exp\left(-\frac{i}{\hbar}Et\right)$$

Hence

$$E\Psi(x, t) = H\psi(x) \exp\left(-\frac{i}{\hbar}Et\right)$$

Cancel exponentials to obtain the time-independent Schrodinger equation

$$E\psi(x) = H\psi(x)$$

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